Evaluation runs: lb (a4-balancer)

generated 2026-11-05 from brain-data a1150b5a2 2026-11-05 14:21:32 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): hash: expected conforms; hash-drop: expected violation; drop: expected violation (a4a is not a README row); hash: expected conforms; hash-drop: expected violation (a4b is not a README row); drop: expected violation; rr: expected violation; rr: expected conforms (vacuous: a client's requests cannot land on two servers); rr: expected conforms; pair-swap-hash: expected conforms (control; m=1 included. Defined for k=2 only: the balancer reads two requests a round); pair-swap-hash: expected conforms (control; m=1 included. Defined for k=2 only: the balancer reads two requests a round); pair-swap-rr: expected conforms (control; m=1 included. Defined for k=2 only: the balancer reads two requests a round); pair-swap-rr: expected violation (as rr; not a README row. Defined for k=2 only); pair-swap-rr: expected conforms (as rr; not a README row. Defined for k=2 only). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{spec}, G_{impl} = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G^{expl}_{spec}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕒 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
1,1,1	1	1	conforms	0.0	4	1 h, 250 GB	1	1	brain03, 2026-11-03
2,1,2	8	2	conforms	0.0	4	1 h, 250 GB	2	9	brain03, 2026-11-03
2,2,2	24	6	conforms	0.0	4	1 h, 250 GB	6	75	brain03, 2026-11-03
3,3,3	45360		🕒 (1 h)			1 h, 250 GB			brain03, 2026-11-03
4,4,4			🕒 (1 h)			1 h, 250 GB			brain03, 2026-11-03

stateless: pair **pair-swap-hash-a4a** against **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
1,1,1	1	1	violation*	0.0	4	1 h, 250 GB	1	1	brain03, 2026-11-03
2,1,2	8	2	conforms	0.0	4	1 h, 250 GB	1	2	brain03, 2026-11-03
2,2,2	24	4	conforms	0.0	4	1 h, 250 GB	1	3	brain03, 2026-11-03
3,3,3	45360	216	violation*	1528.6	12	1 h, 250 GB	216	9797760	brain03, 2026-11-03
4,4,4			🕒 (1 h)			1 h, 250 GB			brain03, 2026-11-03

stateless: pair **pair-swap-rr-a4a** against **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
1,1,1	1	1	violation*	0.0	4	1 h, 250 GB	1	1	brain03, 2026-11-03
2,1,2	8	2	conforms	0.0	4	1 h, 250 GB	2	5	brain03, 2026-11-03
2,2,2	24	4	violation	0.0	4	1 h, 250 GB	4	58	brain03, 2026-11-03
3,3,3	45360	216	violation*	1495.9	13	1 h, 250 GB	216	9797760	brain03, 2026-11-03
4,4,4			🕒 (1 h)			1 h, 250 GB			brain03, 2026-11-03

stateless: pair **rr-a4a** against **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{spec}^{expl}	run
1,1,1	1	1	conforms	0.0	4	1 h, 250 GB	1	1	brain03, 2026-11-03
2,1,2	8	2	conforms	0.0	2	1 h, 250 GB	2	8	brain03, 2026-11-03
2,2,2	24	6	violation	0.0	2	1 h, 250 GB	6	114	brain03, 2026-11-03
3,3,3	45360		(1 h)			1 h, 250 GB			brain03, 2026-11-03

stateless: pair **drop-a4a** against **hash-a4a**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{spec}^{expl}	run
1,1,1	1	1	1	violation	0.0	4	1 h, 250 GB	1	1	brain03, 2026-11-03
2,1,2	8	2	2	violation	0.0	4	1 h, 250 GB	2	16	brain03, 2026-11-03
2,2,2	24	6	6	violation	0.0	4	1 h, 250 GB	6	144	brain03, 2026-11-03
3,3,3	45360			(1 h)			1 h, 250 GB			brain03, 2026-11-03

stateless: pair **hash-drop-a4a** against **hash-a4a**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{spec}^{expl}	run
1,1,1	1	1	1	violation	0.0	4	1 h, 250 GB	1	1	brain03, 2026-11-03
2,1,2	8	2	2	violation	0.0	4	1 h, 250 GB	2	16	brain03, 2026-11-03
2,2,2	24	6	6	violation	0.0	4	1 h, 250 GB	6	144	brain03, 2026-11-03
3,3,3	45360			(1 h)			1 h, 250 GB			brain03, 2026-11-03

stateless: pair **rr-a4a**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{spec}^{expl}	run
1,2,2	2	1	1	violation	0.0	2	1 h, 250 GB	1	2	brain03, 2026-11-03

Not run (stateless). planned but never scheduled (grid points beyond the ladder): drop a4a 1 1 2 fifo, rr a4a 1 1 2 fifo, hash-drop a4a 1 1 2 fifo, pair-swap-rr a4a 1 1 2 fifo, hash a4a 1 1 2 fifo, pair-swap-hash a4a 1 1 2 fifo, hash-drop a4a 1 1 3 fifo, hash a4a 1 1 3 fifo, pair-swap-rr a4a 1 1 3 fifo, pair-swap-hash a4a 1 1 3 fifo, drop a4a 1 1 3 fifo, rr a4a 1 1 3 fifo, hash-drop a4a 1 1 4 fifo, rr a4a 1 1 4 fifo, drop a4a 1 1 4 fifo, pair-swap-hash a4a 1 1 4 fifo, hash a4a 1 1 4 fifo, pair-swap-rr a4a 1 1 4 fifo, hash-drop a4a 1 2 1 fifo, rr a4a 1 2 1 fifo, pair-swap-hash a4a 1 2 1 fifo, hash a4a 1 2 1 fifo, drop a4a 1 2 1 fifo, pair-swap-rr a4a 1 2 1 fifo, … (353 instances in all).

skipped: a smaller size had reached its limit: drop a4a 4 4 4 fifo, hash-drop a4a 4 4 4 fifo, rr a4a 4 4 4 fifo, rr a4a 5 5 5 fifo, pair-swap-hash a4a 5 5 5 fifo, hash-drop a4a 5 5 5 fifo, drop a4a 5 5 5 fifo, pair-swap-rr a4a 5 5 5 fifo, hash a4a 5 5 5 fifo, hash a4a 6 6 6 fifo, pair-swap-rr a4a 6 6 6 fifo, drop a4a 6 6 6 fifo, rr a4a 6 6 6 fifo, pair-swap-hash a4a 6 6 6 fifo, hash-drop a4a 6 6 6 fifo, pair-swap-hash a4a 7 7 7 fifo, drop a4a 7 7 7 fifo, rr a4a 7 7 7 fifo, pair-swap-rr a4a 7 7 7 fifo, hash a4a 7 7 7 fifo, hash-drop a4a 7 7 7 fifo, pair-swap-hash a4a 8 8 8 fifo, hash-drop a4a 8 8 8 fifo, hash a4a 8 8 8 fifo, … (27 instances in all).

stateful

stateful: pair **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
1,1,1	1	1	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-03
2,1,2	8	2	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-03
2,2,2	24	6	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-03
3,3,3	45360	1680	conforms	4.6	2	1 h, 250 GB	brain02, 2026-11-03
4,4,4			(1 h)			1 h, 250 GB	brain02, 2026-11-03

stateful: pair **pair-swap-hash-a4a** against **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
1,1,1	1	1	violation*	0.0	2	1 h, 250 GB	brain02, 2026-11-03
2,1,2	8	2	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-03

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2,2,2	24	4	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-03
3,3,3	45360	216	violation*	4.7	2	1 h, 250 GB	brain02, 2026-11-03
4,4,4			(1 h)			1 h, 250 GB	brain02, 2026-11-03

stateful: pair **pair-swap-rr-a4a** against **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
1,1,1	1	1	violation*	0.0	2	1 h, 250 GB	brain02, 2026-11-03
2,1,2	8	2	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-03
2,2,2	24	4	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
3,3,3	45360	216	violation*	5.8	4	1 h, 250 GB	brain02, 2026-11-03
4,4,4			(1 h)			1 h, 250 GB	brain02, 2026-11-03

stateful: pair **rr-a4a** against **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
1,1,1	1	1	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-03
2,1,2	8	2	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-03
2,2,2	24	6	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-03
3,3,3	45360	1680	violation	5.1	16	1 h, 250 GB	brain02, 2026-11-03
4,4,4			(1 h)			1 h, 250 GB	brain02, 2026-11-03

stateful: pair **drop-a4a** against **hash-a4a**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
1,1,1	1	1	1	violation	2.1	2	1 h, 250 GB	brain02, 2026-11-03
2,1,2	8	2	2	violation	2.1	2	1 h, 250 GB	brain02, 2026-11-03
2,2,2	24	6	6	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
3,3,3	45360	1680	1680	violation	5.0	18	1 h, 250 GB	brain02, 2026-11-03
4,4,4				(1 h)			1 h, 250 GB	brain02, 2026-11-03

stateful: pair **hash-drop-a4a** against **hash-a4a**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
1,1,1	1	1	1	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
2,1,2	8	2	2	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
2,2,2	24	6	6	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
3,3,3	45360	1680	1680	violation	5.3	12	1 h, 250 GB	brain02, 2026-11-03
4,4,4				(1 h)			1 h, 250 GB	brain02, 2026-11-03

stateful: pair **rr-a4a**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
1,2,2	2	1	1	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03

Not run (stateful). planned but never scheduled (grid points beyond the ladder): drop a4a 1 1 2 fifo, rr a4a 1 1 2 fifo, hash-drop a4a 1 1 2 fifo, pair-swap-rr a4a 1 1 2 fifo, hash a4a 1 1 2 fifo, pair-swap-hash a4a 1 1 2 fifo, hash-drop a4a 1 1 3 fifo, hash a4a 1 1 3 fifo, pair-swap-rr a4a 1 1 3 fifo, pair-swap-hash a4a 1 1 3 fifo, drop a4a 1 1 3 fifo, rr a4a 1 1 3 fifo, hash-drop a4a 1 1 4 fifo, rr a4a 1 1 4 fifo, drop a4a 1 1 4 fifo, pair-swap-hash a4a 1 1 4 fifo, hash a4a 1 1 4 fifo, pair-swap-rr a4a 1 1 4 fifo, hash-drop a4a 1 2 1 fifo, rr a4a 1 2 1 fifo, pair-swap-hash a4a 1 2 1 fifo, hash a4a 1 2 1 fifo, drop a4a 1 2 1 fifo, pair-swap-rr a4a 1 2 1 fifo, … (357 instances in all).

skipped: a smaller size had reached its limit: rr a4a 5 5 5 fifo, pair-swap-hash a4a 5 5 5 fifo, hash-drop a4a 5 5 5 fifo, drop a4a 5 5 5 fifo, pair-swap-rr a4a 5 5 5 fifo, hash a4a 5 5 5 fifo, hash a4a 6 6 6 fifo, pair-swap-rr a4a 6 6 6 fifo, drop a4a 6 6 6 fifo, rr a4a 6 6 6 fifo, pair-swap-hash a4a 6 6 6 fifo, hash-drop a4a 6 6 6 fifo, pair-swap-hash a4a 7 7 7 fifo, drop a4a 7 7 7 fifo, rr a4a 7 7 7 fifo, pair-swap-rr a4a 7 7 7 fifo, hash a4a 7 7 7 fifo, hash-drop a4a 7 7 7 fifo, pair-swap-hash a4a 8 8 8 fifo, hash-drop a4a 8 8 8 fifo, hash a4a 8 8 8 fifo, rr a4a 8 8 8 fifo, pair-swap-rr a4a 8 8 8 fifo, drop a4a 8 8 8 fifo.

offline

offline: pair **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	conforms	13.1	4	10 min, 16 GB	0.1	13.1	0.0	in-memory	brain05, 2026-11-02
2,1,2	8	2	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
2,2,2	24	6	conforms	0.2	4	1 h, 250 GB	0.1	0.2	0.0	merge	brain05, 2026-11-03
3,3,3	45360	1680	conforms	7.7	54	4 h, 250 GB	6.2	0.4	1.6	merge	brain05, 2026-11-03

offline: pair **hash-a4a** against **hash-a4b**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	conforms	13.1	4	10 min, 16 GB	0.2	13.1	0.0	in-memory	brain05, 2026-11-02
2,1,2	8	2	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
2,2,2	96	6	conforms	0.2	4	1 h, 250 GB	0.1	0.2	0.0	merge	brain05, 2026-11-03
3,3,3	33067440	1680	conforms	1763.1	40081	1 h, 250 GB	784.1	0.3	979.0	merge	brain05, 2026-11-03

offline: pair **pair-swap-hash-a4a** against **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	violation*	0.1	4	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain05, 2026-11-02
2,1,2	8	2	conforms	0.4	4	10 min, 16 GB	0.1	0.4	0.0	in-memory	brain05, 2026-11-02
2,2,2	24	4	conforms	0.2	4	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain05, 2026-11-02
3,3,3	45360	216	violation*	7.7	54	4 h, 250 GB	6.2	0.1	1.5	merge	brain05, 2026-11-03

offline: pair **pair-swap-hash-a4a** against **hash-a4b**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	violation*	0.2	4	10 min, 16 GB	0.2	0.0	0.0	in-memory	brain05, 2026-11-02
2,1,2	8	2	conforms	0.4	4	10 min, 16 GB	0.1	0.4	0.0	in-memory	brain05, 2026-11-02
2,2,2	96	4	conforms	0.2	4	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain05, 2026-11-02
3,3,3	33067440	216	(1 h)			1 h, 250 GB	784.1	0.1		in-memory	brain05, 2026-11-03

offline: pair **pair-swap-rr-a4a** against **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	violation*	0.4	4	10 min, 16 GB	0.1	0.4	0.0	in-memory	brain05, 2026-11-02
2,1,2	8	2	conforms	0.3	4	10 min, 16 GB	0.1	0.3	0.0	in-memory	brain05, 2026-11-02
2,2,2	24	4	violation	0.2	4	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain05, 2026-11-02
3,3,3	45360	216	violation*	8.0	54	4 h, 250 GB	6.2	0.1	1.8	merge	brain05, 2026-11-03

offline: pair **pair-swap-rr-a4a** against **hash-a4b**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	violation*	0.4	4	10 min, 16 GB	0.2	0.4	0.0	in-memory	brain05, 2026-11-02
2,1,2	8	2	conforms	0.3	4	10 min, 16 GB	0.1	0.3	0.0	in-memory	brain05, 2026-11-02
2,2,2	96	4	conforms	0.2	4	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain05, 2026-11-02
3,3,3	33067440	216	(1 h)			1 h, 250 GB	784.1	0.1		in-memory	brain05, 2026-11-03

offline: pair **rr-a4a** against **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
2,1,2	8	2	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
2,2,2	24	6	violation	0.1	4	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain05, 2026-11-02
3,3,3	45360	1680	violation	7.7	54	4 h, 250 GB	6.2	0.3	1.5	merge	brain05, 2026-11-03

offline: pair **rr-a4a** against **hash-a4b**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	conforms	0.2	4	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
2,1,2	8	2	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
2,2,2	96	6	conforms	0.1	4	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain05, 2026-11-02
3,3,3	33067440	1680	(1 h)			1 h, 250 GB	784.1	0.3		in-memory	brain05, 2026-11-03

offline: pair **drop-a4a** against **hash-a4a**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	1	violation	0.6	4	10 min, 16 GB	0.1	0.6	0.0	in-memory	brain05, 2026-11-02
2,1,2	8	2	2	violation	0.1	4	1 h, 250 GB	0.1	0.1	0.0	merge	brain05, 2026-11-03
2,2,2	24	6	6	violation	0.1	4	1 h, 250 GB	0.1	0.1	0.0	merge	brain05, 2026-11-03
3,3,3	45360	1680	1680	violation	7.8	54	4 h, 250 GB	6.2	0.3	1.6	merge	brain05, 2026-11-03

offline: pair **drop-a4a** against **hash-a4b**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	1	violation	0.6	4	10 min, 16 GB	0.2	0.6	0.0	in-memory	brain05, 2026-11-02
2,1,2	8	2	2	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
2,2,2	96	6	6	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
3,3,3	33067440	1680	1680	violation	1751.5	40081	1 h, 250 GB	784.1	0.3	967.5	merge	brain05, 2026-11-03

offline: pair **hash-drop-a4a** against **hash-a4a**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	1	violation	0.1	4	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain05, 2026-11-02
2,1,2	8	2	2	violation	0.1	4	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain05, 2026-11-02
2,2,2	24	6	6	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
3,3,3	45360	1680	1680	violation	7.7	54	4 h, 250 GB	6.2	0.3	1.5	merge	brain05, 2026-11-03

offline: pair **hash-drop-a4a** against **hash-a4b**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1,1	1	1	1	violation	0.2	4	10 min, 16 GB	0.2	0.0	0.0	in-memory	brain05, 2026-11-02
2,1,2	8	2	2	violation	0.1	4	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain05, 2026-11-02
2,2,2	96	6	6	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
3,3,3	33067440	1680	(1 h)			1 h, 250 GB	784.1	0.3		in-memory	brain05, 2026-11-03

offline: pair **rr-a4a**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,2,2	2	1	1	violation	0.3	4	10 min, 16 GB	0.1	0.3	0.0	in-memory	brain05, 2026-11-02

Not run (offline). planned but never scheduled (grid points beyond the ladder): drop a4a 1 1 2 fifo, rr a4a 1 1 2 fifo, hash-drop a4a 1 1 2 fifo, pair-swap-rr a4a 1 1 2 fifo, hash a4a 1 1 2 fifo, pair-swap-hash a4a 1 1 2 fifo, hash-drop a4a 1 1 3 fifo, hash a4a 1 1 3 fifo, pair-swap-rr a4a 1 1 3 fifo, pair-swap-hash a4a 1 1 3 fifo, drop a4a 1 1 3 fifo, rr a4a 1 1 3 fifo, hash-drop a4a 1 1 4 fifo, rr a4a 1 1 4 fifo, drop a4a 1 1 4 fifo, pair-swap-hash a4a 1 1 4 fifo, hash a4a 1 1 4 fifo, pair-swap-rr a4a 1 1 4 fifo, hash-drop a4a 1 2 1 fifo, rr a4a 1 2 1 fifo, pair-swap-hash a4a 1 2 1 fifo, hash a4a 1 2 1 fifo, drop a4a 1 2 1 fifo, pair-swap-rr a4a 1 2 1 fifo, … (368 instances in all).

refused (an input missing at run time): pair-swap-hash a4a 4 4 4 fifo, pair-swap-rr a4a 4 4 4 fifo, hash a4a 4 4 4 fifo, pair-swap-rr a4a 5 5 5 fifo, pair-swap-hash a4a 5 5 5 fifo.

skipped: a smaller size had reached its limit: drop a4a 4 4 4 fifo, hash-drop a4a 4 4 4 fifo, rr a4a 4 4 4 fifo, rr a4a 5 5 5 fifo, hash-drop a4a 5 5 5 fifo, drop a4a 5 5 5 fifo, hash a4a 5 5 5 fifo, hash a4a 6 6 6 fifo, pair-swap-rr a4a 6 6 6 fifo, drop a4a 6 6 6 fifo, rr a4a 6 6 6 fifo, pair-swap-hash a4a 6 6 6 fifo, hash-drop a4a 6 6 6 fifo, pair-swap-hash a4a 7 7 7 fifo, drop a4a 7 7 7 fifo, rr a4a 7 7 7 fifo, pair-swap-rr a4a 7 7 7 fifo, hash a4a 7 7 7 fifo, hash-drop a4a 7 7 7 fifo, pair-swap-hash a4a 8 8 8 fifo, hash-drop a4a 8 8 8 fifo, hash a4a 8 8 8 fifo, rr a4a 8 8 8 fifo, pair-swap-rr a4a 8 8 8 fifo, … (25 instances in all).

cross online, Java implementation (JMC) against the Rust specification export

cross online, Java implementation (JMC) against the Rust specification export: pair **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	conforms	0.6	146	1 h, 250 GB		brain05, 2026-11-02
2,1,2	8	2	conforms	0.6	150	1 h, 250 GB		brain05, 2026-11-02
2,2,2	24	6	conforms	0.7	152	1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **hash-a4a** against **hash-a4b**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	conforms	0.6	146	1 h, 250 GB		brain05, 2026-11-02
2,1,2	8	2	conforms	0.6	152	1 h, 250 GB		brain05, 2026-11-02
2,2,2	96	6	conforms	0.7	156	1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **pair-swap-hash-a4a** against **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	violation*	0.6	148	1 h, 250 GB	43	brain05, 2026-11-02
2,1,2	8	2	conforms	0.6	148	1 h, 250 GB		brain05, 2026-11-02
2,2,2	24	4	conforms	0.7	152	1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **pair-swap-hash-a4a** against **hash-a4b**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	violation*	0.6	156	1 h, 250 GB	42	brain05, 2026-11-02
2,1,2	8	2	conforms	0.6	150	1 h, 250 GB		brain05, 2026-11-02
2,2,2	96	4	conforms	0.7	156	1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **pair-swap-rr-a4a** against **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	violation*	0.5	146	1 h, 250 GB	31	brain05, 2026-11-02
2,1,2	8	2	conforms	0.6	148	1 h, 250 GB		brain05, 2026-11-02

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2,2	24	4	violation	0.6	148	1 h, 250 GB	86	brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair pair-swap-rr-a4a against hash-a4b, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	violation*	0.6	147	1 h, 250 GB	35	brain05, 2026-11-02
2,1,2	8	2	conforms	0.6	148	1 h, 250 GB		brain05, 2026-11-02
2,2,2	96	4	conforms	0.6	154	1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair rr-a4a against hash-a4a, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	conforms	0.6	150	1 h, 250 GB		brain05, 2026-11-02
2,1,2	8	2	conforms	0.6	144	1 h, 250 GB		brain05, 2026-11-02
2,2,2	24	6	violation	0.7	148	1 h, 250 GB	83	brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair rr-a4a against hash-a4b, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	conforms	0.6	148	1 h, 250 GB		brain05, 2026-11-02
2,1,2	8	2	conforms	0.6	150	1 h, 250 GB		brain05, 2026-11-02
2,2,2	96	6	conforms	0.7	156	1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair drop-a4a against hash-a4a, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	1	violation	0.6	144	1 h, 250 GB	38	brain05, 2026-11-02
2,1,2	8	2	2	violation	0.6	144	1 h, 250 GB	45	brain05, 2026-11-02
2,2,2	24	6	6	violation	0.7	156	1 h, 250 GB	55	brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair drop-a4a against hash-a4b, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	1	violation	0.6	154	1 h, 250 GB	36	brain05, 2026-11-02
2,1,2	8	2	2	violation	0.6	146	1 h, 250 GB	43	brain05, 2026-11-02
2,2,2	96	6	6	violation	0.7	174	1 h, 250 GB	53	brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair hash-drop-a4a against hash-a4a, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	1	violation	0.6	152	1 h, 250 GB	39	brain05, 2026-11-02
2,1,2	8	2	2	violation	0.6	177	1 h, 250 GB	45	brain05, 2026-11-02
2,2,2	24	6	6	violation	0.7	148	1 h, 250 GB	52	brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **hash-drop-a4a** against **hash-a4b**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	1	violation	0.6	158	1 h, 250 GB	39	brain05, 2026-11-02
2,1,2	8	2	2	violation	0.6	148	1 h, 250 GB	38	brain05, 2026-11-02
2,2,2	96	6	6	violation	0.7	152	1 h, 250 GB	44	brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **rr-a4a**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1,2,2	2	1	1	violation	0.6	147	1 h, 250 GB	43	brain05, 2026-11-02

Not run (cross online, Java implementation (JMC) against the Rust specification export). planned but never scheduled (grid points beyond the ladder): pair-swap-rr a4a 1 1 1 fifo, hash a4a 1 1 1 fifo, drop a4a 1 1 1 fifo, rr a4a 1 1 1 fifo, pair-swap-hash a4a 1 1 1 fifo, hash-drop a4a 1 1 1 fifo, rr a4a 1 2 2 fifo, drop a4a 2 1 2 fifo, hash-drop a4a 2 1 2 fifo, hash a4a 2 1 2 fifo, pair-swap-hash a4a 2 1 2 fifo, pair-swap-rr a4a 2 1 2 fifo, rr a4a 2 1 2 fifo, pair-swap-rr a4a 2 2 2 fifo, rr a4a 2 2 2 fifo, hash-drop a4a 2 2 2 fifo, hash a4a 2 2 2 fifo, drop a4a 2 2 2 fifo, pair-swap-hash a4a 2 2 2 fifo.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01
2,1,2	8	2	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01
2,2,2	24	6	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair **hash-a4a** against **hash-a4b**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01
2,1,2	8	2	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01
2,2,2	96	6	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair **pair-swap-hash-a4a** against **hash-a4a**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	violation*	0.0	4	1 h, 250 GB	1	brain04, 2026-11-01
2,1,2	8	2	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01
2,2,2	24	4	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair **pair-swap-hash-a4a** against **hash-a4b**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	violation*	0.0	4	1 h, 250 GB	1	brain04, 2026-11-01
2,1,2	8	2	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2,2	96	4	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair pair-swap-rr-a4a against hash-a4a, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	violation*	0.0	4	1 h, 250 GB	1	brain04, 2026-11-01
2,1,2	8	2	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01
2,2,2	24	4	violation	0.0	4	1 h, 250 GB	2	brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair pair-swap-rr-a4a against hash-a4b, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	violation*	0.0	4	1 h, 250 GB	1	brain04, 2026-11-01
2,1,2	8	2	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01
2,2,2	96	4	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair rr-a4a against hash-a4a, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01
2,1,2	8	2	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01
2,2,2	24	6	violation	0.0	4	1 h, 250 GB	2	brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair rr-a4a against hash-a4b, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01
2,1,2	8	2	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01
2,2,2	96	6	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair drop-a4a against hash-a4a, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	1	violation	0.0	2	1 h, 250 GB	2	brain04, 2026-11-01
2,1,2	8	2	2	violation	0.0	4	1 h, 250 GB	1	brain04, 2026-11-01
2,2,2	24	6	6	violation	0.0	4	1 h, 250 GB	2	brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair drop-a4a against hash-a4b, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	1	violation	0.0	4	1 h, 250 GB	2	brain04, 2026-11-01
2,1,2	8	2	2	violation	0.0	4	1 h, 250 GB	2	brain04, 2026-11-01
2,2,2	96	6	6	violation	0.0	4	1 h, 250 GB	2	brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair hash-drop-a4a against hash-a4a, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	1	violation	0.0	4	1 h, 250 GB	1	brain04, 2026-11-01
2,1,2	8	2	2	violation	0.0	4	1 h, 250 GB	1	brain04, 2026-11-01
2,2,2	24	6	6	violation	0.0	4	1 h, 250 GB	2	brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair hash-drop-a4a against hash-a4b, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1,1,1	1	1	1	violation	0.0	4	1 h, 250 GB	1	brain04, 2026-11-01
2,1,2	8	2	2	violation	0.0	4	1 h, 250 GB	2	brain04, 2026-11-01
2,2,2	96	6	6	violation	0.0	4	1 h, 250 GB	2	brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair rr-a4a, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1,2,2	2	1	1	violation	0.0	4	1 h, 250 GB	1	brain04, 2026-11-01

Not run (cross online, Rust implementation (TraceForge) against the Java specification export). left out by the table rules (ok): pair-swap-rr a4a 1 1 1 fifo, hash a4a 1 1 1 fifo, drop a4a 1 1 1 fifo, rr a4a 1 1 1 fifo, pair-swap-hash a4a 1 1 1 fifo, hash-drop a4a 1 1 1 fifo, rr a4a 1 2 2 fifo, drop a4a 2 1 2 fifo, hash-drop a4a 2 1 2 fifo, hash a4a 2 1 2 fifo, pair-swap-hash a4a 2 1 2 fifo, pair-swap-rr a4a 2 1 2 fifo, rr a4a 2 1 2 fifo, pair-swap-rr a4a 2 2 2 fifo, rr a4a 2 2 2 fifo, hash-drop a4a 2 2 2 fifo, hash a4a 2 2 2 fifo, drop a4a 2 2 2 fifo, pair-swap-hash a4a 2 2 2 fifo.